

Face Attendance

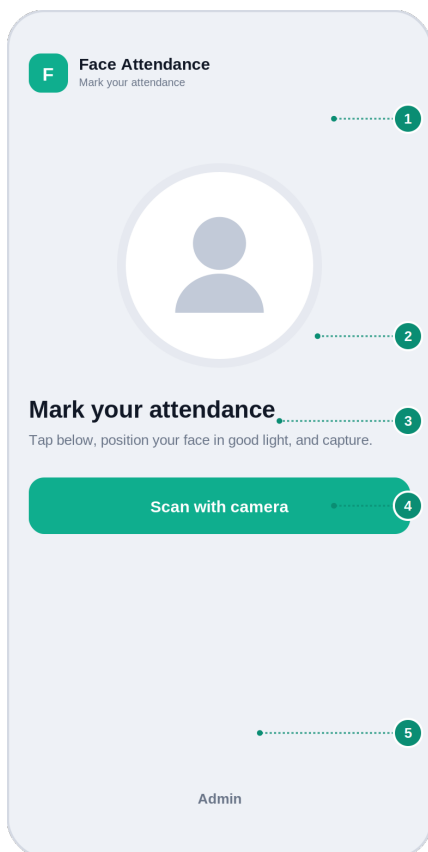
User Guide — annotated screen-by-screen walkthrough

An offline face-recognition attendance app. Face detection, liveness, and recognition run entirely on the device; recorded attendance uploads automatically once the device is back online and is then purged from local storage.

Each page shows one screen with **numbered markers** on the interface, a key explaining each marker, and an “under the hood” note on the feature behind it.

Model: w600k_mbf_float16.tflite • React Native (Android 8+/iOS 12+) • Works fully offline

Screen images are faithful UI mockups rendered from the application's actual styles and on-screen text.



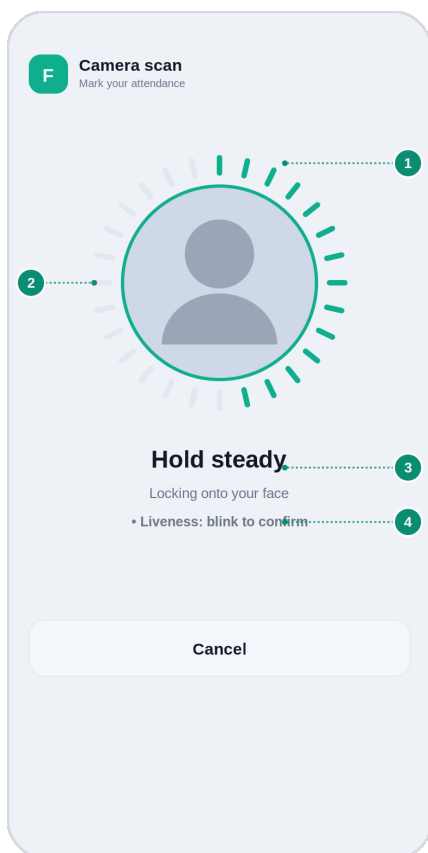
1. Home

First screen when the app opens.

1. **App header** — brand and current mode.
2. **Face placeholder** — the live camera appears here once a scan starts.
3. **Instruction** — guidance to position the face in good light.
4. **Scan with camera** — the single primary action; starts marking attendance.
5. **Admin** — PIN-protected entry to enrollment and sync tools.

Under the hood

The home screen exposes exactly one user action so anyone can mark attendance without training. Recognition does not start until the user taps Scan, keeping the camera off until needed.



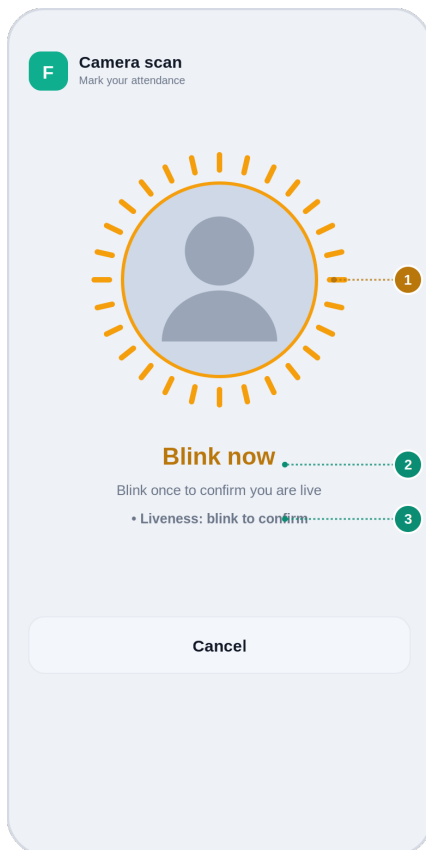
2. Scanning — locking on

Phase 1: finding and holding the face.

1. **Live camera** in the scan ring.
2. **Teal ring fills** as a steady, centered, close face is detected.
3. **“Hold steady”** headline while locking.
4. **Liveness hint** — blink check is coming next.

Under the hood

Phase 1 uses fast on-device face detection on lightweight preview frames — it only locates the face and checks size/steadiness. No image is matched yet, which keeps this stage responsive.



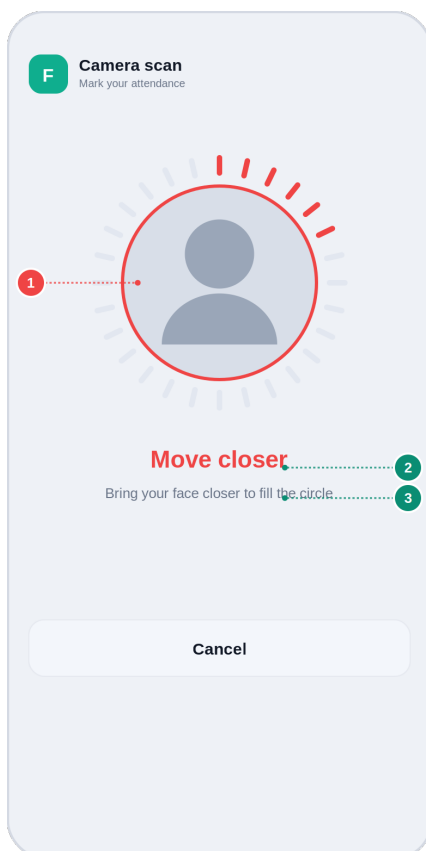
3. Liveness — blink

Phase 1 continued: anti-spoofing.

1. Ring turns **amber** once the face is locked.
2. “**Blink now**” — the liveness prompt.
3. **Liveness tracker** confirms a live person.

How blink liveness defeats fraud

A printed photo or a face shown on a screen cannot perform a natural blink. By requiring an open→closed→open eye transition, the app rejects photo and screen replay attacks — the offline anti-spoofing measure the system is built around.



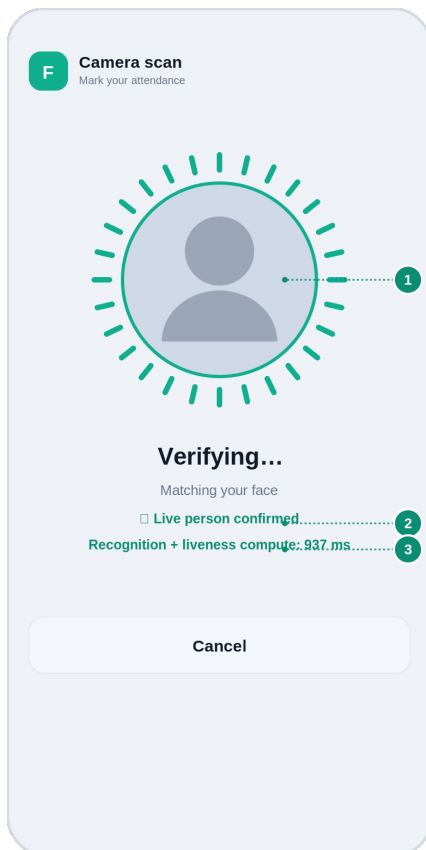
4. “Move closer” guidance

Shown when the face is too far.

1. Ring turns **red** when the face is too small in frame.
2. “**Move closer**” headline.
3. **Helper text** — fill the circle with your face.

Why proximity matters

Recognition accuracy depends on a large, clear face crop. The app enforces a minimum face size before it will capture, so a too-distant face is corrected up front rather than producing a weak, unreliable match.



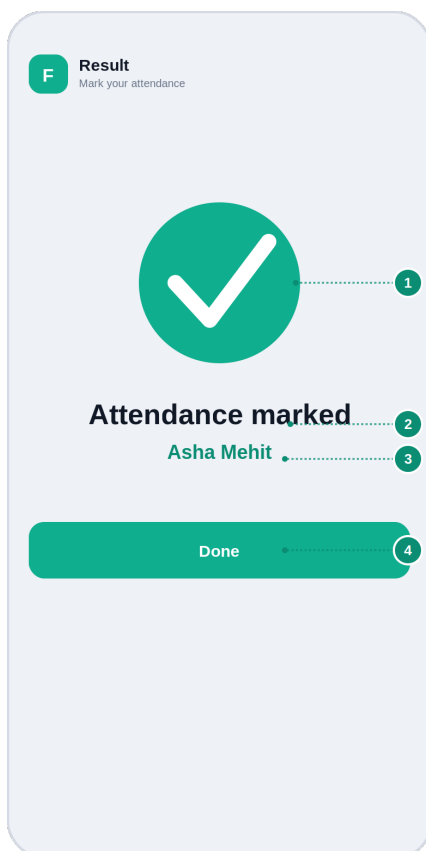
5. Recognition

Phase 2: the on-device face match.

1. **Full teal ring** — capturing and matching.
2. **“Live person confirmed”** after the blink.
3. **Compute timing** — recognition + liveness in milliseconds.

Sub-second, fully offline

One frame is captured, the face is aligned to a fixed geometry, and a 512-dimension embedding is produced by w600k_mbf_float16.tflite and compared to enrolled faces by cosine similarity. The whole match runs on the device in well under a second — no network involved.



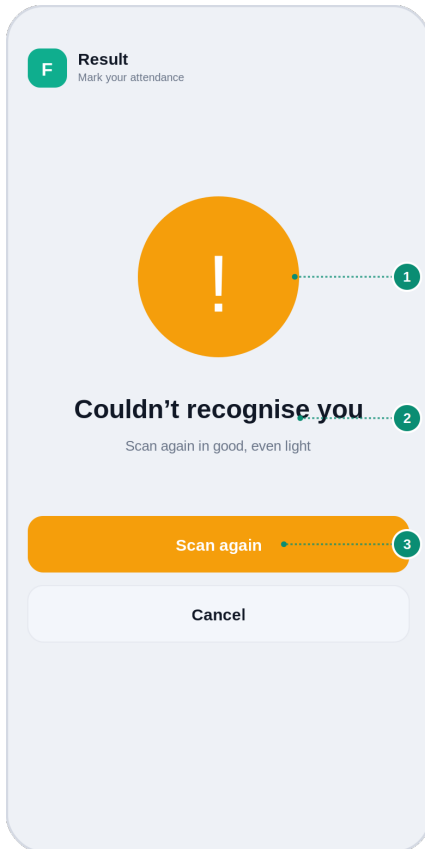
6. Attendance marked

Success result.

1. **Teal check** — confident match.
2. **“Attendance marked”** confirmation.
3. **Recognized name** shown to the user.
4. **Done** — return home for the next person.

What happens to the record

The attendance entry (name, person id, timestamp) is written to a local queue on the device. Nothing else is required from the user — upload happens automatically later when the device is online.



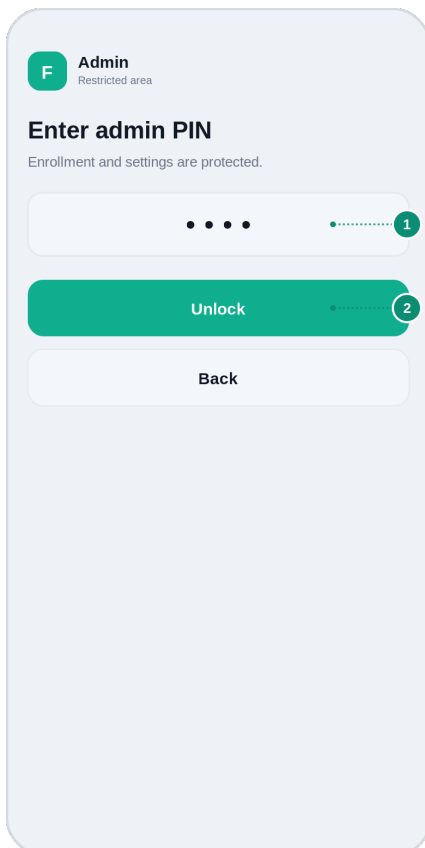
7. Not recognized

No confident match found.

1. **Amber alert** — no match.
2. **“Couldn't recognise you”** message.
3. **Scan again** — retry with better framing/light.

Common fixes

Most retries succeed with even, front-on lighting, removing sunglasses or a mask, and moving so the face fills the ring. A person who is not enrolled will never match — enroll them first via Admin.



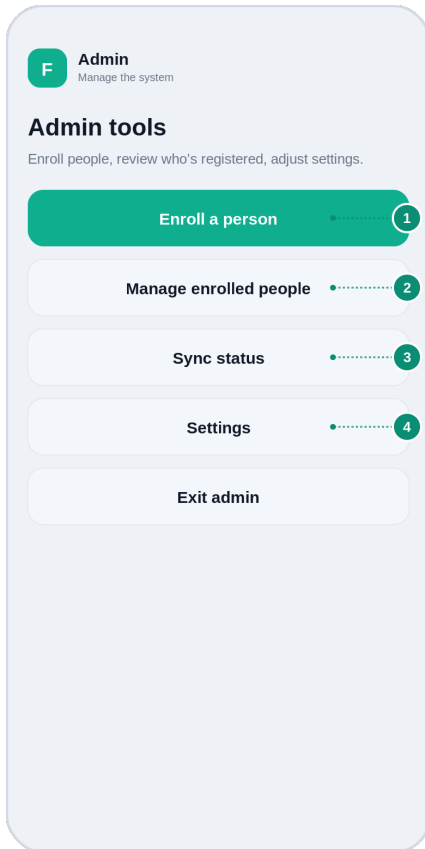
8. Admin — PIN

Gate protecting admin tools.

1. **PIN field** — masked entry.
2. **Unlock** — an incorrect PIN highlights amber and clears.

Why it is gated

Enrollment, people management, and settings sit behind a PIN so ordinary users can only mark attendance. This separates the high-trust setup actions from everyday use.



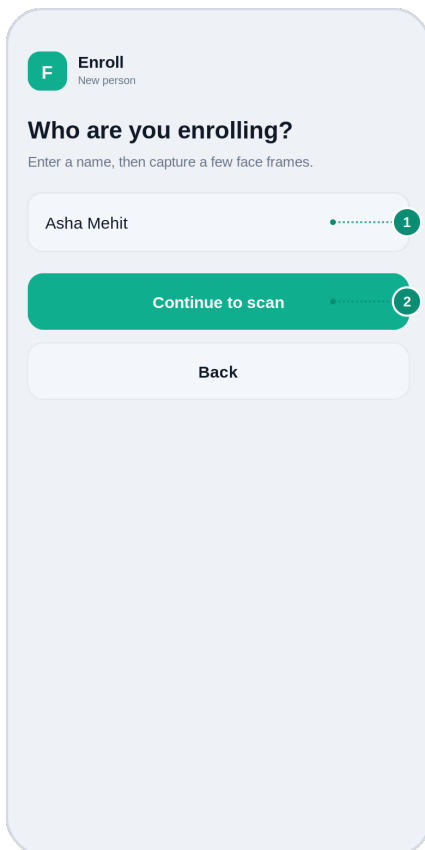
9. Admin — tools

The admin landing screen.

1. **Enroll a person** — register a new face.
2. **Manage enrolled people** — review/remove.
3. **Sync status** — pending/synced counts and manual sync.
4. **Settings** — thresholds and configuration.

Under the hood

Enrollment is the only path to register a face profile. All admin functions are device-local; no admin action requires connectivity.



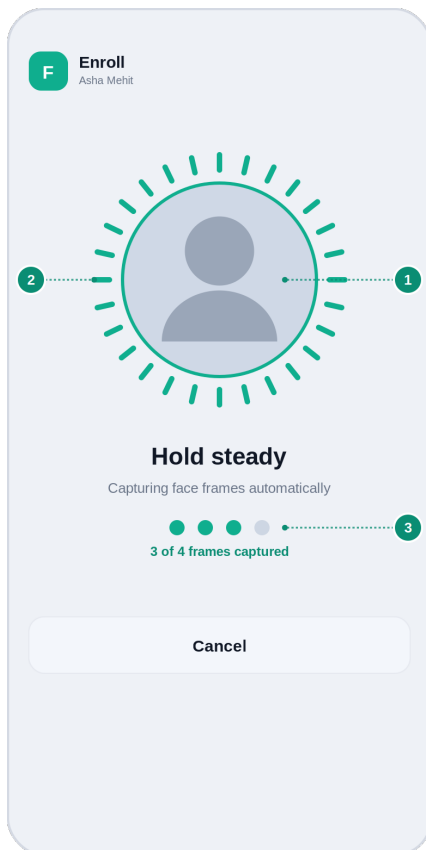
10. Enroll — Step 1: name

Registering a new face (1 of 2).

1. **Name field** — who is being enrolled.
2. **Continue to scan** — runs the guided face capture.

How a profile is built

The guided scan captures a few frames and averages them into one compact 512-dimension face profile, stored on the device only. Good enrollment light and a neutral pose make later recognition more reliable.



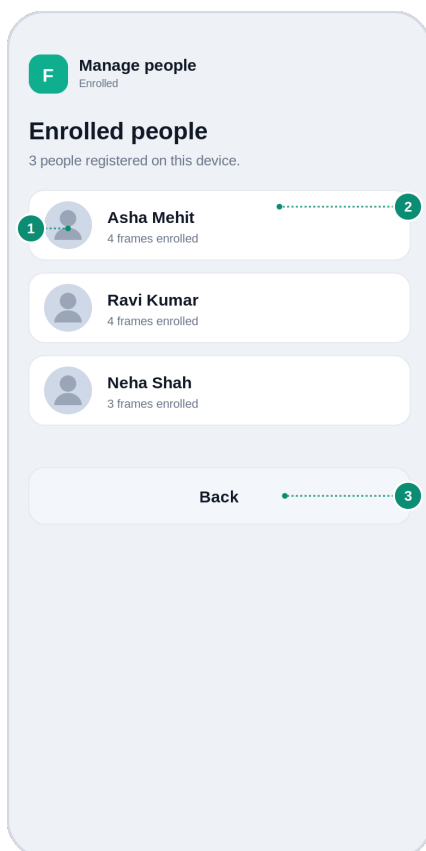
11. Enroll — Step 2: capture the face

Recording the face (2 of 2).

1. **Live camera** — the person looks into the ring.
2. **Teal ring locks** on a steady, centered face.
3. **Frames captured automatically** — several are taken and averaged into one profile.

How the face profile is built

Enrollment reuses the same guided scan as attendance but without the blink step. Once the face is locked, the app captures several frames and averages their embeddings into a single, robust 512-dimension profile, stored on the device only. More frames in varied light make later recognition more reliable.



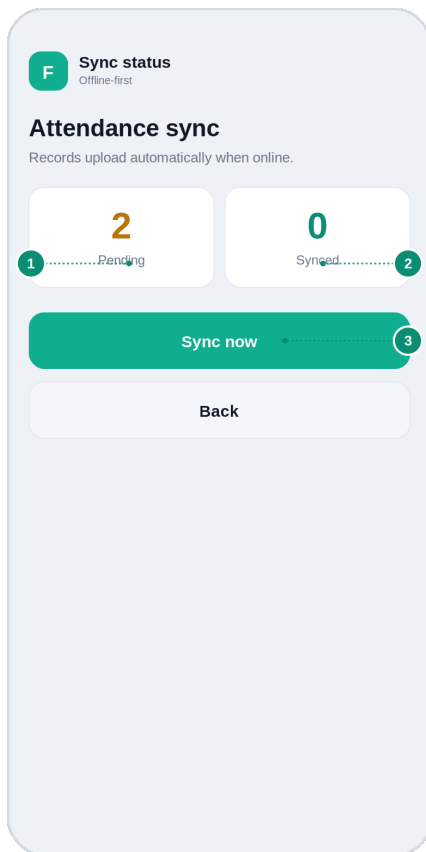
12. Manage enrolled people

Reviewing who is registered.

1. **Person thumbnail** from enrollment.
2. **Name & frame count** per person.
3. **Back** — return to admin tools.

Local-only data

Enrolled face profiles never leave the device. Removing a person deletes their profile immediately from local storage.



13. Sync status

Offline-first upload and purge.

1. **Pending count** — records not yet uploaded.
2. **Synced count** — records confirmed by the server.
3. **Sync now** — upload immediately on demand.

Sync & purge mechanism

Attendance is stored locally first, then uploaded in batches when connectivity returns (automatically, or via Sync now). Once the server confirms receipt, the synced records are purged from the device — so no device retains attendance data after it has been uploaded.

The whole flow, in one minute

Tap **Scan with camera** → the app locks onto the face, asks for a **blink** to confirm a live person, then **matches the face on-device** in under a second. A successful match shows “**Attendance marked**” with the name, is saved to a local queue, and is **uploaded automatically** when the device is online — after which the local copy is purged.

Everything that identifies a face runs offline

The only network activity in the entire app is uploading attendance records (name, id, timestamp) to your configured server. Face detection, the blink liveness check, and recognition all run locally on the phone.